

# Shang Liu

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## EDUCATION

### **Ph.D. in Operations and Analytics**

Business School, Imperial College London (ICL), London, UK.

*Sept. 2022 – Jun. 2027 (Expected)*

Advisor: Xiaocheng Li

### **M.Res. in Operations and Analytics (Distinction)**

Business School, Imperial College London (ICL), London, UK.

*Sept. 2021 – Jun. 2022*

### **B.S. in Mathematics and Applied Mathematics**

School of Mathematical Sciences, Peking University (PKU), Beijing, China.

*Sept. 2016 – Jun. 2021*

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## RESEARCH INTERESTS

My research focuses on enhancing the alignment, robustness, and ethical deployment of large language models (LLMs) through interdisciplinary frameworks of human-AI collaboration, game theory, statistical learning, and optimization. I aim to develop both theoretically guaranteed (e.g., convergence analyses, complexity bounds) and empirically valid methodologies. My research interests also lie in the development and theoretical analysis of machine learning models, with a particular focus on the role of uncertainty, including uncertainty quantification, active learning, and human-AI interaction.

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## PUBLICATIONS

(reverse chronological order; \*: equal contribution)

### **Wasserstein Distributionally Robust Regret Optimization for Reinforcement Learning from Human Feedback**

Yikai Wang\*, Shang Liu\*, Jose Blanchet, [arXiv](#)

### **Incentivizing High-Quality Human Annotations with Golden Questions**

Shang Liu, Zhongze Cai, Hanzhao Wang, Zhongyao Ma, Xiaocheng Li, [arXiv](#)

### **How Humans Help LLMs: Assessing and Incentivizing Human Preference Annotators**

Shang Liu\*, Hanzhao Wang\*, Zhongyao Ma, Xiaocheng Li, *under Minor Revision at Management Science*, [arXiv](#)

### **Reward Modeling with Ordinal Feedback: Wisdom of the Crowd**

Shang Liu\*, Yu Pan\*, Guanting Chen, Xiaocheng Li, *ICML 2025*, [arXiv](#)

**Towards Better Understanding of In-Context Learning Ability from In-Context Uncertainty Quantification**  
Shang Liu\*, Zhongze Cai\*, Guanting Chen, Xiaocheng Li, *Transactions on Machine Learning Research*, [arXiv](#)

**Towards Better Statistical Understanding of Watermarking LLMs**

Zhongze Cai\*, Shang Liu\*, Hanzhao Wang\*, Huaiyang Zhong, Xiaocheng Li, *accepted at Journal of the American Statistical Association*, [arXiv](#)

**When No-Rejection Learning is Consistent for Regression with Rejection**

Xiaocheng Li, Shang Liu, Chunlin Sun, Hanzhao Wang, *AISTATS 2024*, [arXiv](#)

**Understanding Uncertainty Sampling via Equivalent Loss**

Shang Liu, Xiaocheng Li, [arXiv](#)

**Distribution-Free Model-Agnostic Regression Calibration via Nonparametric Methods**

Shang Liu\*, Zhongze Cai\*, Xiaocheng Li, *NeurIPS 2023*, [arXiv](#)

**Maximum Optimality Margin: A Unified Approach for Contextual Linear Programming and Inverse Linear Programming**

Chunlin Sun\*, Shang Liu\*, Xiaocheng Li, *ICML 2023*, [arXiv](#)

**Non-stationary Bandits with Knapsacks**

Shang Liu, Jiashuo Jiang, Xiaocheng Li, *NeurIPS 2022*, [arXiv](#)

**Online Bin Packing with Known  $T$**

Shang Liu, Xiaocheng Li, *under Major Revision at Mathematics of Operations Research*, [arXiv](#)

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## **PRIZES & AWARDS**

Dean's List for Academic Excellence at Imperial College Business School, 2021/2022

Outstanding Graduate of Peking University, 2021

Wusi Scholarships at Peking University, 2016/2017, 2017/2018

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## **SEVICE**

Reviewer: NeurIPS, ICLR, AISTATS, ICML, TMLR, INFORMS IJDS

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## **TEACHING**

TA:

Machine Learning, Imperial College London, Spring 2024/2025/2026

Generative AI and Large Language Models, Imperial College London, Spring 2025/2026